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ORIGINAL ARTICLE

# The *pncA* gene mutations of *Mycobacterium tuberculosis* in multidrug-resistant tuberculosis

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**Abstract**

The *pncA* gene encodes pyrazinamidase enzyme which converts drug pyrazinamide to active form pyrazinoic acid, but mutations in this gene can prevent enzyme activity which leads to pyrazinamide resistance. The cross-sectional study was carried out during 2016–2017 for 12 months. The purpose of the study was to detect mutation at codon 12 and codon 85 in the *pncA* gene in local multidrug-resistant tuberculosis (MDR-TB) patients by developing a simple molecular test so that disease could be detected timely in the local population. DNA extracted from sputum-cultured samples from MDR-TB patients and subjected to semi-multiplex allele-specific PCR by using self-designed primers against the *pncA* gene. Among 75 samples, 53 samples were subjected to molecular analysis based on purified DNA quantity and quality. The primers produced 250 and 480 bp fragments, indicating the mutations at codon 12 (aspartate to alanine) and codon 85 (leucine to proline) respectively. MDR-TB was more common in the age group 21–40 years. Fifty-seven percent of samples ($n = 30$) were found positive for *pncA* mutations, whereas 43% of samples ($n = 23$) showed negative results. Thirteen percent of samples ($n = 4$) had mutations at codon 12 in which aspartate was converted to alanine, and they produced an amplified product of 480 bp. Eighty-seven percent of samples ($n = 26$) had mutations at codon 85 in

**Abbreviations:** DNA, deoxyribonucleic acid (It is the hereditary material in humans and almost all other organisms. The information in DNA is stored as a code made up of four chemical bases: adenine (A), guanine (G), cytosine (C), and thymine (T)); XDR-TB, Extensively drug-resistant tuberculosis (It is a type of multidrug-resistant tuberculosis that is resistant to isoniazid and rifampin, plus any fluoroquinolone and at least one of three injectable second-line drugs i.e., amikacin, kanamycin, or capreomycin); LCD-array, low-cost density array (it is designed and tested towards its specificity and sensitivity for five genera causing tick-borne diseases); MGIT, mycobacteria growth indicator tube (It was introduced for the growth and detection of mycobacteria from clinical specimens); MDR-TB, multidrug-resistant tuberculosis (It does not respond to at least anti-TB drugs-isoniazid and rifampicin); NCBI BLAST, National Center for Biotechnology Information The Basic Local Alignment Search Tool (BLAST discovers the regions of similarity between biological sequences); PCR, polymerase chain reaction (It's a test to detect genetic material from a specific organism, such as a virus) pncA; PZAse, Pyrazinamidase (Pyrazinamidase activity has been found to correlate with pyrazinamide sensitivity in strains of Mycobacterium tuberculosis); PZA, Pyrazinamide (Pyrazinamide is a medication used to treat tuberculosis); POA, Pyrazinamide into Pyrazinoic acid (Pyrazinamide is a prodrug that is converted into the active form pyrazinoic acid (POA) by bacterial nicotinamidase/pyrazinamidase (PZase)).; SNPs, Single nucleotide polymorphisms (They are a type of polymorphism involving variation of a single base pair); TAE Buffer, Tris-acetate-EDTA (TAE Buffer (It is used for electrophoresis of nucleic acids in agarose and polyacrylamide gels); TB, Tuberculosis (It is a serious infectious disease that mainly affects the lungs caused by bacteria called Mycobacterium tuberculosis).

which leucine was converted to proline and amplified product size was 250 bp. The mutations were simple nucleotide substitutions. The prevalence of mutations in which leucine was substituted by proline was higher than the mutations in which aspartate was substituted by alanine. A high prevalence of substitution mutation (CTG → CCG; leucine to proline) was detected in MDR-TB cases. Earlier detection of MDR-TB via an effective molecular diagnostic method can control the MDR tuberculosis spread in the population.

**KEYWORDS**
leucine, multidrug-resistant tuberculosis, proline, pyrazinamide

## 1 | INTRODUCTION

Tuberculosis (TB) because of *Mycobacterium tuberculosis* complex members remains a typical human irresistible infection.[1] TB is one of the greatest health challenges which the world is confronting and is the second significant reason for mortality, especially in poor and low monetary nations.[2] Even with the current advances in medication, TB is still a noteworthy reason for deaths at an expansive scale around the world.[3,4] Around 8.6 million individuals suffered from TB all over the world in 2012.[5] TB has been viewed as a worldwide general health issue since 1993, and certain measures have been undertaken to control this infection.[6,7] Every year around 430,000 individuals, including 15,000 children suffer from TB in Pakistan, and TB is a cause of the death of nearly 70,000 people. Pakistan's position is at the sixth place universally among the 22 high TB trouble nations.[8] Pakistan is evaluated to have the fourth most prevalent nation of multidrug resistant TB (MDR-TB) internationally.[5] As indicated by a report, roughly 51% of the cases were recorded from Punjab, 23% of the cases were recorded in Sindh, 15% of the cases were recorded in Khyber Pakhtunkhwa, and 3.5% of the cases were recorded from Baluchistan and Azad Kashmir.[9]

Treatment of TB includes a 6-month regimen comprising isoniazid, rifampicin, and pyrazinamide (PZA) in which PZA is given for 2 months and afterward isoniazid and rifampin are given for 4 months. This is the favored treatment for patients with completely susceptible bacteria who hold fast to treatment.[10] The principal first-line drugs for the treatment of TB comprise rifampicin, isoniazid, ethambutol, streptomycin, and PZA.[11] These drugs used for the treatment of TB have various efficacies. These drugs possess bactericidal activity. MDR-TB is a kind of TB that is resistant to at least isoniazid and rifampicin.[12] In Pakistan, around 35% of previously diagnosed patients have developed MDR-TB.[13] The major reason for MDR-TB is an incomplete course of relevant antibiotic dose, failure to complete the treatment therapy, and exposure to MDR strain.[14] Extensively drug-resistant TB (XDR-TB) is a type of TB, which shows resistance to at least four of the major TB drugs. The two most intense first-line anti-TB drugs, isoniazid and rifampicin along with resistance to any fluoroquinolones (e.g., levofloxacin or moxifloxacin) and second-line drugs: amikacin, capreomycin, or kanamycin. An XDR-TB causes more morbidity and mortality than any other TB.[15]

PZA is an essential first-line drug for TB. It is also part of a short-course treatment regimen known as DOTS (directly observed therapy short course) along with rifampin, isoniazid, and ethambutol.[16,17] PZA is a structural analog of nicotinamide.[18] PZA is considered a prodrug that gains entry into the tubercle bacilli through passive diffusion followed by its hydrolysis by an enzyme known as pyrazinamidase (PZAse) that transforms PZA into pyrazinoic acid (POA). POA is protonated to H+ -POA when it is exported outside the *M. tuberculosis* and again re-enters into bacteria where H+ is released inside while the POA is pumped towards the exterior. This process is cyclic, and its repetition enhances the acidification inside the *M. tuberculosis*, resulting in deadly effects. [16,19,20] The *pncA* gene of *M. tuberculosis* encodes a PZAse enzyme. It is 561-nucleotide long. Several studies have reported mutation-like insertions and deletions spread along the entire length of the *pncA* gene, and its promoter region is a reason behind the loss of PZAse.[16,19,21–24] PZA is an important first-line drug that is biologically inactive and is converted into its active form POA by the action of enzyme PZAse that is encoded by the *pncA* gene, but due to mutation in the *pncA* gene, PZAse activity is lost, which results in MDR-TB. The present study is aimed to find mutations in the *pncA* gene of *M. ,tuberculosis*. Semi-multiplex allele-specific PCR is a quick and economical method to detect resistance-related genetic mutations simultaneously against the first line as well as second-line TB drugs.[25]

Bamaga et al.[26] reported that PZAse was produced from *pncA* genes in mycobacteria. In *M. tuberculosis*, the hydrolysis of PZA occurs by the action of PZAse, resulting in the production of a bactericidal compound named POA. This acid kills tubercle bacilli. Nucleotide sequencing of *pncA* genes present in mycobacteria was carried out to assure the significance of sequence variability. The multiplex PCR assay has been performed for the amplification of the pncA region in seven clinical mycobacteria and initiated by designing three sets of primers. Each species generated bands in conjunction with the activity of PZAse illustrated that species of mycobacteria were different from each other and could be easily identified. Xu et al.[27] used about 493 isolates of *M. tuberculosis* from China in five field settings to investigate PZA resistance by the *pncA* gene through gene sequencing. Transmission of PZA resistance isolates plays a vital role in the development of PZA-resistant TB. The results demonstrated that conducting PZA susceptibility tests in multidrug resistant isolates is vital and therefore a modification in the treatment should be done accordingly. Zignol et al.[28] proposed that fluoroquinolones and PZA are crucial anti-TB drugs in new rifampicin-saving regimens. A very little data at the population level about the degree of resistance to these medications are accessible. Molecular epidemiology analysis has been carried out, and an investigation was done through the population-based surveys from Pakistan, Bangladesh, Azerbaijan, Belarus, and South Africa. PZA and fluoroquinolones resistances were checked in TB patients. PZA resistance has been performed through the sequencing of the *pncA* gene, whereas the MGIT (mycobacteria growth indicator tube) system was used to determine the susceptibility of fluoroquinolones. In 4972 patients, the PZA resistance was evaluated. The resistance level fluctuated considerably in the overview settings (3.0–42.1%). All settings were found with PZA resistance in association with rifampicin resistance. Doustdar et al.[29] reported that 70.58% of samples had mutations in the *pncA* gene. Jonmalung et al. [30] demonstrated 6% PZA resistance in susceptible isolates and 49% PZA resistance in MDR-TB isolates. The present study was aimed to look for mutations in the *pncA* gene of the *M. tuberculosis* at codon positions 12 and 85 in local MDR-TB patients for early diagnosis of disease. As mutation at codon 85 (CTG → CCG; leucine to proline) is present in the majority of cases so it can be used for early detection of MDR-TB cases along with other genomic targets.

## 2 | MATERIALS AND METHODS

### 2.1 | Sample collection and processing

A total of 75 sputum-cultured samples from MDR-TB patients with a history of resistance against first-line anti-TB drugs were obtained from Gulab Devi Chest Hospital, Lahore, and University of Health Sciences, Lahore, Pakistan, during 2016–2017 for the current cross-sectional study. Among 75 samples, 53 samples were subjected to molecular analysis based on purified DNA quantity and quality. Informed consent was obtained from all patients or their attendants. The patient's medical history was recorded, and samples were analyzed for mutations in the *pncA* gene after isolating genomic DNA. Samples were taken in autoclaved Eppendorf tubes in sterile water suspension (0.5 mL). The samples were then stored at −10°C for further use after heat inactivation.

### 2.2 | Genomic DNA isolation

Following reagents were used: RL buffer (washing buffer), AL buffer (cell lysis buffer), PP buffer (protein precipitate buffer), isopropanol/absolute ethanol, and proteinase K for the genomic DNA isolation. The total genomic DNA from sputum cultured samples was isolated by following the procedure of Gene All Genomic DNA isolation kit (Tiangen Biotech) with some modifications.

### 2.3 | DNA quantification and quality

Isolated DNA was subjected to UV (ultraviolet) spectrophotometry for quantification. The DNA dilution was appropriately prepared, and the optical density was calculated at 260 and 280 nm. Electrophoresis was performed to check the quality of DNA. Following reagents were used to check the quality of isolated DNA: agarose, 1 kb DNA ladder, 6× DNA gel loading dye, 50× TAE (Tris-acetate-EDTA) buffer, and ethidium bromide.

### 2.4 | Semimultiplex allele-specific PCR of the *pncA* gene

#### 2.4.1 | Primer designing

In silico studies were performed to design primers. The PCR-amplified products of the *pncA* gene were determined by using self-designed primers. Gene search in the NCBI database was carried out, and then the accession number of the gene was noted. The primers were designed by using the accession number U59967. Then BLAST analysis of the gene was performed. As mentioned in various pieces of literature, the gene sequence was translated to find the location of the mutation in the *pncA* gene. The mutations were located at positions 12 and 85 of amino acids. Then the expected product size was determined and calculated. By using the oligonucleotide properties calculator, parameters like GC (guanine-cytosine) content, melting temperature, self-annealing of primers were evaluated. Table 1 illustrates the primers and their sequences. To check the mutation in codon 12, the GAC forward primer was designed as [5′

**TABLE 1** Sequence of primers for *pncA* gene amplification

| Primers | Sequence | Primer length nucleotide |
|---|---|---|
| pncA forward primer (1) | 5′ CGT CGA CGT GCA GAA CGA 3′ | 18 |
| pncA forward primer (2) | 5′ GGA CTT CCA TCC CAG TCT 3′ | 18 |
| pncA reverse primer | 5′ AGC ACC CTG GTG GCC AA 3′ | 17 |

**TABLE 2** Tuberculosis Prevalence in different age groups of people

| Serial no. | Age group(years) | Male | Female | Overall | Percentage |
|---|---|---|---|---|---|
| 1 | Less than 20 | 1 | 14 | 15 | 20 |
| 2 | 21–40 | 12 | 29 | 41 | 55 |
| 3 | 41–60 | 7 | 7 | 14 | 19 |
| 4 | 60 above | 2 | 3 | 5 | 6% |

CGT CGA CGT GCA GAA CGA 3′]. To check the mutation in codon 85, the CTG forward primer was designed as [5′ GGA CTT CCA TCC CAG TCT 3′]. The reverse primer for both forward primers was designed as [5′ AGC ACC CTG GTG GCC AA 3′].

#### 2.4.2 | PCR amplification

The PCR amplification was carried out by using designed primers. The PCR-amplified products of the *pncA* gene were determined by using two forward primers and one common reverse primer. The primers produced 250 and 480 bp fragments, indicating the mutations at codon 12 (aspartate to alanine) and codon 85 (leucine to proline), respectively. The semimultiplex allele-specific PCR condition of primers was optimized on DNA isolated from positive cultures of *M. tuberculosis* showing multidrug resistance isolated from MDR-TB patients.

The following procedure was followed: The master mixture was prepared by mixing the PCR reagents. After the preparation of the master mixture, 43.4 $\mu$L of the master mixture was pipetted into the reaction tube for each polymerase chain reaction. A 2.5-$\mu$L of DNA extract, Taq DNA polymerase 0.5 $\mu$L, and primers (1.2 each) were added into the PCR tube and mixed with 43.4 $\mu$L of Master mixture. After that, all reaction tubes were transferred into the PCR system to perform PCR amplification.

#### 2.4.3 | PCR reaction cycling conditions

The melting temperature for the *pncA* gene was optimized at 59°C. After 40 cycles, the reaction terminated at 4°C. The PCR-amplified product was verified by agarose gel electrophoresis. Following reagents were used: agarose, 1 kb DNA ladder, 6× DNA gel loading dye, 50× TAE buffer, and ethidium bromide.

## 3 | RESULTS

### 3.1 | Age and gender of patients

The 30% samples were from male patients and 70% were from female patients. MDR-tuberculosis prevalence is found higher in females than males. In the age group 1, 20% patients were found less than 20 years; in group 2, patients were between 20 and 40 years (55%); in group 3, patients were between 41 and 60 years (19%); and in group 4, patients were found above 60 years (6%). The prevalence of TB was higher in patients that belong to group 2 (20-40 years) and was low in patients with age above 60 years (Table 2).

### 3.2 | DNA quantity and purity analysis of *M. tuberculosis*

The Gene All Genomic DNA kit was used for the extraction of DNA from *M. tuberculosis* in the sputum-cultured samples. The analysis of the isolated DNA was carried out by taking the absorbance at 260 nm, and DNA quantity was found in the range of 30–50 ng/$\mu$L. The 260/280 ratio of absorbance was found in the range of 1.6–2.0, indicating good quality DNA, free of proteins contamination with a DNA band of above 10 kb was observed on the agarose gel. Good quality DNA was obtained from 53 samples out of a total of 75 samples. The good quality DNA was further used for *pncA* gene mutational analysis through semi-multiplex allele-specific PCR.

### 3.3 | Mutational analysis of the *pncA* gene by the semi-multiplex allele-specific PCR

Mutational analysis of the *pncA* gene was carried out by further following semi-multiplex allele-specific PCR. The

**TABLE 3** Results for *pncA* mutations

| Number of tested samples | Positive *pncA* mutations | Negative *pncA* mutations |
|---|---|---|
| 53 | 30 (57%) | 23 (43%) |

primers GAC forward primer and CTG forward primer were designed to target mutations in the *pnc*A gene at codons 12 and 85, respectively. The mutations that were targeted were single nucleotide substitutions in which the aspartate was replaced by alanine at position 12 of amino acid and leucine was converted to proline at position 85 of the amino acid. Out of 53 samples, 30 samples were positive for *pncA* mutations (57%), whereas no mutation was found in the *pncA* gene in 23 samples (43%). Results for the *pncA* mutations are shown in Table 3.

## 3.4 | Frequency of targeted mutations in the *pncA* gene

There were two targeted mutations in the *pncA* gene. One mutation was at position 12 of amino acid in which aspartate was substituted by alanine, and the other mutation was at position 85 in which leucine was substituted by proline. Amplicons of 250 and 480 bp were generated by the semi-multiplex allele-specific PCR. The primers produced 250 and 480 bp fragments, indicating the mutations at codon 12 (aspartate to alanine) and codon 85 (leucine to proline), respectively. These amplified products were analyzed by 1.5% gel electrophoresis. The results of gel electrophoresis were documented on a gel documentation system, which is shown in Figures 1–4.

Among 30 positive *pncA* mutations in the samples, four samples (13%) had mutations in the *pncA* gene on the 12th amino acid and 26 samples (87%) had mutations in the *pncA* gene on the 85th amino acid. Therefore, from these results, it was concluded that mutations at the 85th position were greater than the mutations at the 12th position of the amino acids in the *pncA* gene. The frequency of the *pncA* mutations is given in Table 4. The type of mutation in the *pncA* gene along with a fragment size is given in Table 5.

## 3.5 | Bioinformatics

In silico studies were performed for the primer designing. The following sequence of the *pncA* gene was obtained under accession number: U59967. The designed primers produced 100% similarity in the NCBI BLAST sequence with the *pncA* gene. The sequence of the pncA gene is given in Figure 5. The region highlighted as green indicates

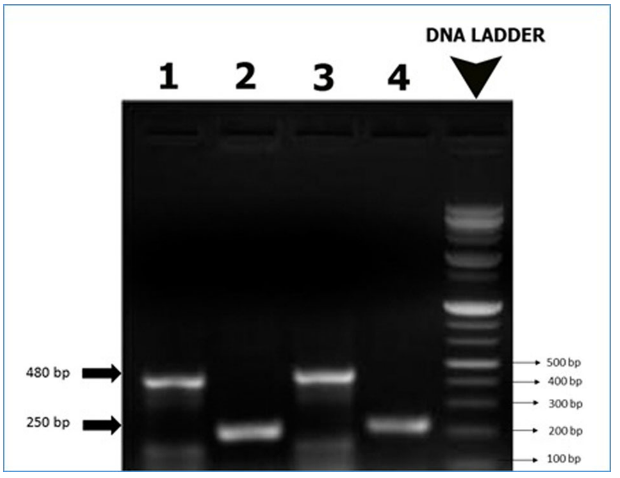


**FIG 1** Representative photograph showing semi-multiplex PCR analysis of the *pncA* gene of *M. tuberculosis* from MDR-TB patients on 1.5% agarose gel electrophoresis. Lanes 1 and 3: Samples showing PCR amplicons of 480 bp fragments and Lanes 2 and 4: Samples showing PCR amplicons of 250 bp fragments of the *pncA* gene. DNA ladder of 1 kb (+) (cat. #DM03, Enzynomics). The 250 and 480 bp fragments indicating the mutations at codon 12 (aspartate to alanine) and codon 85 ([eucine to proline), respectively, in the *pncA* gene

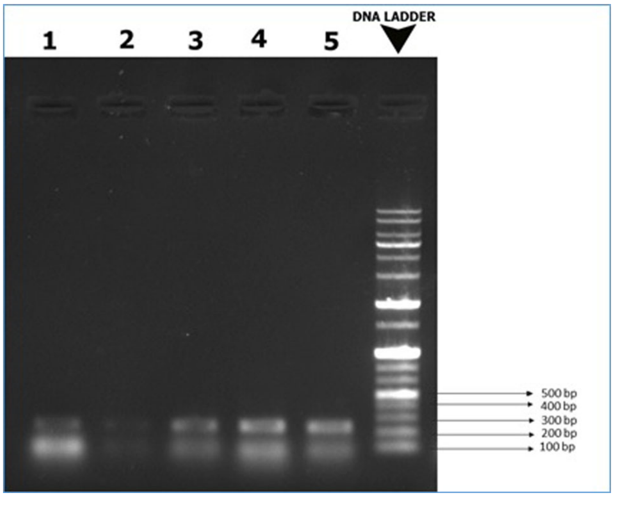


**FIG 2** Second representative photograph showing semi-multiplex PCR analysis of the *pncA* gene of *M. tuberculosis* from MDR-TB patients on 1.5% agarose gel electrophoresis. Lanes 1–5: Samples showing 250 bp fragments of the *pncA* gene, a DNA ladder of 1 kb (+) (cat. #DM03, Enzynomics). The 250 and 480 bp fragments indicating the mutations at codon 12 (aspartate to alanine) and codon 85 (leucine to proline), respectively, in the *pncA* gene

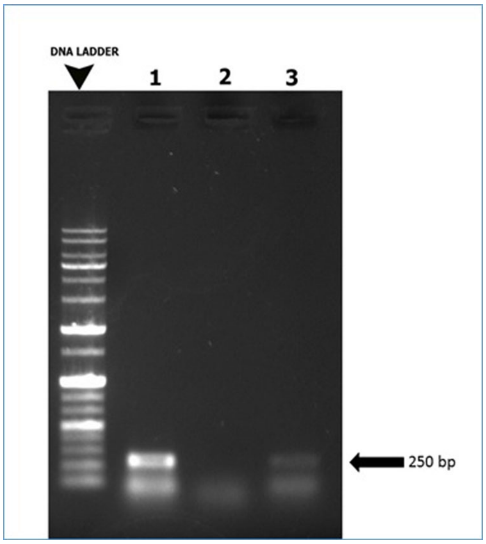


**FIG 3** Third representative photograph showing semi-multiplex PCR analysis of the *pncA gene* of *M. tuberculosis* from MDR-TB patients on 1.5% agarose gel electrophoresis. Lanes 1, 3: Patient DNA samples representing amplified fragments of 250 bp size of the *pncA* gene; lane 2: negative control; M: DNA ladder of 1 kb (+) (Cat. No. DM03, Enzynomics). The 250 and 480 bp fragments indicating the mutations at codon 12 (aspartate to alanine) and at codon 85 (leucine to proline), respectively, in the *pncA* gene

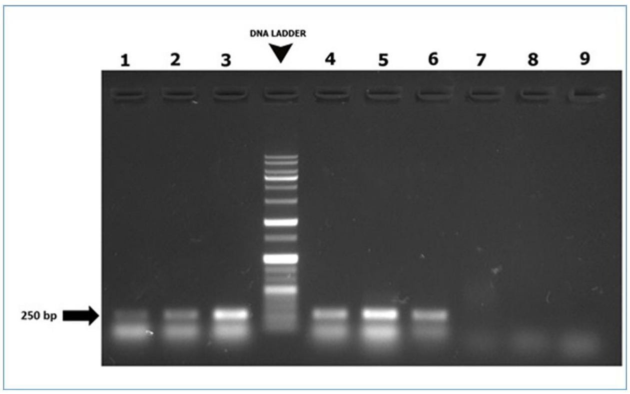


**FIG 4** Fourth representative photograph showing semi-multiplex PCR analysis of the *pncA* gene of *M. tuberculosis* from MDR-TB patients on 1.5% agarose gel electrophoresis. Lanes 1–6: MDR-TB samples showed 250 bp amplified fragment; Lane 7: negative control; Lanes 8, 9: MDR-TB samples negative for *pncA* mutations; M: DNA ladder of 1 kb (+) (Cat. No. DM03, Enzynomics). The 250 and 480 bp fragments indicating the mutations at codon 12 (aspartate to alanine) and at codon 85 (leucine to proline), respectively, in the *pncA* gene

**TABLE 4** Frequency of *pncA* mutations

| *Pnca* Mutations | Frequency | Percentage |
|---|---|---|
| At codon 12 | 4 | 13 |
| At codon 85 | 26 | 87 |

the position of forward primer GAC and pink indicates the position of forward primer CTG, respectively. The region highlighted in blue indicates the position of the reverse primer.

## 4 | DISCUSSION

In the current research, the mutations in the *pncA* gene at codons 12 and 85 were detected by simplified semi-multiplex allele-specific PCR by using self-designed primers. We reported the absence and presence of mutations in the *pncA* gene at respective codon positions. Fifty-seven percent of samples ($n = 30$) were found positive for *pncA* mutations, while 43% ($n = 23$) showed negative results. Among mutation-positive samples, 13% ($n = 4$) samples had mutations at codon 12 in which aspartate was converted to alanine and they produced an amplified product of 480 bp while 87% ($n = 26$) samples had mutations at codon 85 in which leucine was converted to proline and the amplified product size was 250 bp. As mutations in the *pncA* gene prevent PZAse enzyme activity, hence the conversion of the drug PZA to an active form POA is stopped and leads to PZA resistance in *M. tuberculosis*. This drug-resistant *M. tuberculosis* causes MDR-TB, which if not detected and treated in time in MDR-TB patients then may result in serious health consequences. These mutations in the *pncA* gene indicated that the patients will respond to PZA drug or not. If mutations are present, then TB is resistant to PZA and if no mutation is present then PZA can be used in the treatment. The presence of MDR-TB was more prevalent in the age group 21–40 years. The results of current research illustrated that mutations were simple nucleotide substitutions. The prevalence of mutations in which leucine was substituted by proline was higher than the mutations in which aspartate was substituted by alanine. In some previous studies, the substitution in which leucine was converted to proline at codon 85 was more frequent than the mutation in which aspartate was converted to alanine at codon 12.[21,31,32]

**TABLE 5** The type of mutation and amino acid changes in the pncA gene and fragment size of the bands appeared

| *pncA* Mutations | Fragmentsize | Type of mutation | Amino acid change | Nucleotide change |
|---|---|---|---|---|
| At codon 12 | 480 bp | Substitution | Asp → Ala | GAC → GCC |
| At codon 85 | 250 bp | Substitution | Leu → Pro | CTG → CCG |

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atgcgggcgttgatcatcgtcgacgtgcagaacgacttctgcgagggtggctcgctggcg
 M  R  A  L  I  I  V  D  V  Q  N  D  F  C  E  G  G  S  L  A
gtaaccggtggcgccgcgctggcccgcgccatcagcgactacctggccgaagcggcggac
 V  T  G  G  A  A  L  A  R  A  I  S  D  Y  L  A  E  A  A  D
taccatcacgtcgtggcaaccaaggacttccacatcgacccgggtgaccacttctccggc
 Y  H  H  V  V  A  T  K  D  F  H  I  D  P  G  D  H  F  S  G
acaccggactattcctcgtcgtggccaccgcattgcgtcagcggtactcccggcgcggac
 T  P  D  Y  S  S  S  W  P  P  H  C  V  S  G  T  P  G  A  D
ttccatcccagtctggacacgtcggcaatcgaggcggtgttctacaagggtgcctacacc
 F  H  P  S  L  D  T  S  A  I  E  A  V  F  Y  K  G  A  Y  T
ggagcgtacagcggcttcgaaggagtcgacgagaacggcacgccactgctgaattggctg
 G  A  Y  S  G  F  E  G  V  D  E  N  G  T  P  L  L  N  W  L
cggcaacgcggcgtcgatgaggtcgatgtggtcggtattgccaccgatcattgtgtgcgc
 R  Q  R  G  V  D  E  V  D  V  V  G  I  A  T  D  H  C  V  R
cagacggccgaggacgcggtacgcaatggcttggccaccagggtgctggtggacctgaca
 Q  T  A  E  D  A  V  R  N  G  L  A  T  R  V  L  V  D  L  T
gcgggtgtgtcggccgataccaccgtcgccgcgctggaggagatgcgcaccgccagcgtc
 A  G  V  S  A  D  T  T  V  A  A  L  E  E  M  R  T  A  S  V
gagttggtttgcagctcctga
 E  L  V  C  S  S  -
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**FIG 5** The sequence of the *pncA* gene (accession no. U59967). The region highlighted as green indicates the position of forward primer GAC and pink indicates the position of forward primer CTG, respectively. The region highlighted in blue indicates the position of the reverse primer

Cheng et al.[31] clarified the genetic premise of PZA resistance by doing sequencing the *pncA* gene in 59 PZA -resistant clinical isolates got from Korea, Canada, and United States. He described that out of 59, 53 PZA -resistant strains had *pncA* mutations, showing that *pncA* mutations assume an essential part in PZA resistance. Strain typing showed that these strains share practically indistinguishable IS6110 patterns. The data firmly showed the spread of a PZA -resistant strain. Miyagi et al.[33] worked on 26 PZA-resistant isolates obtained from Japan and found out the association between mutations in the *pncA* gene and PZA resistance. Out of 26 PZA-resistant isolates, five isolates were positive for PZAse activity and 21 showed negative PZAse activity. Among these, 20 isolates had different *pncA* mutations, whereas one negative isolate showed no mutation on the *pncA* gene. Barco et al.[32] inspected the region of mutations in the *pncA* gene in live strains and 31 PZA-resistant *M. tuberculosis* strains. Out of the 31 isolates that were resistant to PZA, 26 (83.9%) had mutations in the *pncA* gene. Juréen et al.[34] reported that the sensitive strains showed no mutations except some silent mutations, whereas *pncA* mutations were observed in all resistant strains except one. Zhang et al.[35] reported that out of 47 PZA-resistant isolates, 44 had *pncA* mutations that occurred generally near PZase's active or metal ion–binding sites, showed no PZAse activity while three of them carried wild-type pncA sequence by maintaining PZAse activity. Ando et al.[36] found that 53% of isolates were PZA-resistant. All these isolates lack PZAse activity and comprise at least one mutation in the *pncA* gene. It was concluded that a major part of multidrug-resistant *M. tuberculosis* isolates was resistant to PZA in Japan. Pandey et al.[37] reported 11 isolates particularly those a with multidrug resistance phenotype had mutations. Muthaiah et al.[38] explored the treatment collapse of pulmonary TB in south India. It was found that 79.5% of samples were restricted to three regions of pncA.

Hong et al.[39] demonstrated *pncA* gene characteristics in MDR-tuberculosis isolates along with its correlation to the PZA resistance. For all the isolates, *pncA* sequencing along with PZAse activity testing was carried out. The types of *pncA* mutations along with their frequency were determined. Correlation analysis was performed for PZAse activity, *pncA* mutation, and PZA susceptibility. Out of 127 isolates, 62 (48.8%) were resistant to PZA. Out of these 62 PZA-resistant isolates, about 45 isolates showed negative PZAse activity and possessed other *pncA* mutations. The rate of mutation was 77.4% among all PZA-resistant isolates. Sheen et al.[40] investigated that the mutations in the *pncA* gene encoding PZAse are considered the primary reason for the PZA resistance in *M. tuberculosis*. Zenteno-Cuevas et al.[41] reported that out of 127 drug-resistant isolates obtained, 42 (33%) were PZA -resistant strains. PZA was changed over to POA by the action of the PZAse enzyme that is encoded by the gene *pncA* present in *M. tuberculosis*. PZA resistance can be rapidly diagnosed by molecular identification.[42] Miotto et al.[42] identified 280 genetic variants. Aono et al.[43] worked on seven different isolates that possessed differential PZA susceptibility and performed tests on the colonies of these isolates. Pang et al.[44] showed that resistance in PZA that was determined at pH 5.7 (94.9%) in the liquid culture demonstrated a better correlation with the genetic modifications of multidrug-resistant *M. tuberculosis* isolates.

PZA is one of the four frontline TB treatments available during the first 2 months of therapy. The significant frequency of PZA resistance, particularly among those with MDR-TB, highlights the need for pharmaceutical regimens that may be employed in patients with PZA-resistant MDR-TB. In our study, the prevalence of mutations in which leucine was substituted by proline was higher than the mutations in which aspartate was substituted by alanine. A high prevalence of substitution mutation (CTG → CCG; leucine to proline) was detected in MDR-TB cases. Moradi et al.[45] reported that only one MDR-TB isolate resistant to PZA (14.3%) had a mutation in the *pncA* gene, and no mutations were found in the other PZA-resistant isolates. Resistance to PZA in other PZA-resistant strains might be due to mutations in other genes or a variety of other reasons.[45] Atashi et al.[46] used the GeneXpert MTB/RIF test and the conventional proportional technique to identify rifampicin resistance in new cases of TB in the west and northwest Iran. Mohajeri et al.[47] employed a low-cost density array (LCD-array) to identify changes in the 90-bp rpoB region. They concluded that rifampicin resistance determining region–related site mutations in rifampicin-resistant *M. tuberculosis* isolates may be detected using the microarray rpoB.[47] Mohajeri et al.[48] used the multiplex PCR technique, and 10% of isolates were identified as Beijing genotypes, whereas the remaining 90% isolates were identified as non-Beijing genotypes. Drug-resistant isolates were found in three of the cases. They affirmed that the multiplex-PCR technique is practical, reliable, and capable of distinguishing mixed illnesses. To distinguish mixed infections, spoligotyping should be used in addition to the multiplex PCR technique.[48]

MDR-TB must be diagnosed quickly so that effective second-line medication may be started quickly, improving treatment outcomes and limiting MDR-TB transmission. Several molecular techniques have been developed to help in the fast identification of mutations linked to MDR-TB.[49] Sadri et al.[49] used a molecular test to evaluate the prevalence of significant drug resistance mutations across the katG and inhA loci of multidrug-resistant-MTB isolates. PZA genotypic testing may be a more straightforward way to determine medication susceptibility. However, there is a scarcity of information on the prevalence and diversity of *pncA* mutations in clinical settings. Even though PZA is widely used to treat TB, medication susceptibility testing is not regularly accessible. Simple assays for PZA susceptibility testing are needed; however, they may be challenging to design due to genotypic or phenotypic complexity. Because of the possible interaction of PZA with new TB medicines, regular PZA testing will become more important and helpful.[50] In Uganda, we discovered four variants linked to PZA phenotypic resistance: K96R, T142R, R154G, and V180F. In Uganda, phenotypic PZA resistance is common among TB patients. The poor sensitivity of *pncA* gene sequencing validates previously reported discrepancies, suggesting that PZA resistance in Mycobacterium TB is caused by other processes.[51] The first time, the *pncA* gene alterations in PZA-resistant isolates from Mexico were described by Cuevas-Córdoba et al. in 2017.[52] PZA resistance was found in 42 of the 127 drug-resistant isolates collected. In 34 isolates, the pncA sequences revealed 26 changes: 18 SNPs, 4 insertions, and 4 deletions. In eight sequences/isolates, no changes were found. The mutations L120P and K96R were the two most common. For the first time, 12 modifications were discovered and reported. Authors inferred that the wide range of pncA alterations implies the presence of a significant number of PZA - resistant bacteria in the area.[52]

## 5 | CONCLUSION

In MDR-TB patients, the percentage prevalence of mutation involving the *pncA* gene in which leucine was substituted by proline (codon 85) was found higher (87%) than the mutation in which aspartate was substituted by alanine at codon 12 (13%). In the current research, the mutations in the *pncA* gene were detected by sizes of amplified fragments in a simplified molecular method, which gave an idea about the absence and presence of mutations in the *pncA* gene at specific positions and hence early detection of MDR-TB. By effective molecular diagnostic methods, the spread and transmission of MDR tuberculosis can be reduced in the population.

### 5.1 | Limitations

The study can be carried out further on a relatively larger number of MDR-TB patients in the future in different geographical regions to determine the prevalence of mutations at codon 85 and 12 in the *pncA* gene along with the effectiveness of the described molecular method.

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## CONFLICTS OF INTEREST

The authors report no conflict of interests.

## ETHICAL APPROVAL

The research proposal was approved by the ethical committee of Lahore Garrison University, Lahore, Pakistan. All authors have given consent to participate in the study.

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